RA23403 – MOTION CONTROL SYSTEMS

AIR FLOW BASED BALL LEVITATION SYSTEM

Submitted By

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BONAFIDE CERTIFICATE

Certified that this project report **"DESIGN, MODELLING, SIMULATION, AND CONTROL OF A BALL LEVITATION SYSTEM USING AIR FLOW"** is the Bonafide word of **Angadh Roy K (2024511001), Abdiel Zion S (2024511006) and Madhesh BS (2024511017)** who carried out the project work under my supervision.

SIGNATURE **SIGNATURE**

HEAD OF THE DEPARTMENT **SUPERVISOR**

ABSTRACT:

This report presents the design and control of an airflow-based ball levitation system in which a 45 mm ball is suspended inside a transparent tube using aerodynamic lift generated by a blower fan. The system represents a classic unstable control problem requiring continuous feedback to maintain equilibrium against gravitational forces. The project integrates sensing, actuation, embedded control and real-time feedback using an Arduino Uno microcontroller. An HC-SR04 ultrasonic sensor provides continuous position measurement of the ball within the 300 mm tube of 55 mm inner diameter, while an L298N motor driver regulates blower fan speed through pulse-width modulation. A PID-based control strategy is implemented to stabilize the ball at a desired levitation height by dynamically adjusting airflow velocity in response to positional error. The system is powered by a 12.4 V, 2000 mAh LiPo battery, enabling portable and self-contained operation. The project demonstrates key concepts in nonlinear control including aerodynamic instability, closed-loop feedback regulation, actuator dynamics and real-time digital control implementation on embedded hardware.

INTRODUCTION:

The design and implementation of an airflow-based ball levitation system represent a sophisticated intersection of fluid dynamics, electronic interfacing, and control theory. The primary aim of this project is to engineer a system capable of suspending a 45 mm ball inside a 300 mm transparent tube by developing a robust feedback control loop centred around an Arduino Uno microcontroller. By implementing a Proportional-Integral-Derivative (PID) controller, the project seeks to transform an inherently unstable aerodynamic environment into a stable, controlled state. This process requires the seamless integration of sensing, actuation, and real-time processing to create a working prototype that allows for a detailed evaluation of system stability and dynamic response behaviour.

The core challenge lies in the fact that aerodynamic ball levitation is a benchmark problem in control engineering due to its natural instability. Without active intervention, a suspended ball will either fall under gravity when airflow is insufficient or accelerate toward the top of the tube when fan speed is excessive. The relationship between fan speed and the resulting lift force is inherently nonlinear, making precise control a non-trivial task. The significance of this work extends beyond the simple act of levitation, as it serves as a practical application of closed-loop feedback systems and the integration of embedded electronics with complex control algorithms in a real physical environment.

The system architecture is composed of four critical subsystems that work in a synchronized loop to achieve its goals. First, a sensing subsystem employing an HC-SR04 ultrasonic sensor continuously monitors the vertical position of the ball within the tube, providing the necessary feedback data. This data is fed into the Arduino Uno controller, which compares the current position against a desired setpoint and calculates the necessary corrective adjustment using PID computation. To execute this adjustment, the controller sends a Pulse Width Modulation (PWM) signal to an L298N motor driver, which regulates the voltage supplied to the blower fan actuator. The blower fan then generates the precise airflow velocity required to counteract gravity and maintain the ball at the target height. The entire system is powered by a 12.4 V, 2000 mAh LiPo battery, enabling self-contained and portable operation.

The development of this report was supported by a thorough review of design schematics, simulation models, and experimental observations, ensuring that the hardware specifications and PID tuning insights are accurately reflected in the

final prototype. Through this integrated approach, the project demonstrates how theoretical control principles drawn from classical feedback theory can be successfully applied to solve the physical constraints of an unstable aerodynamic system.

SOLIDWORKS DESIGN:

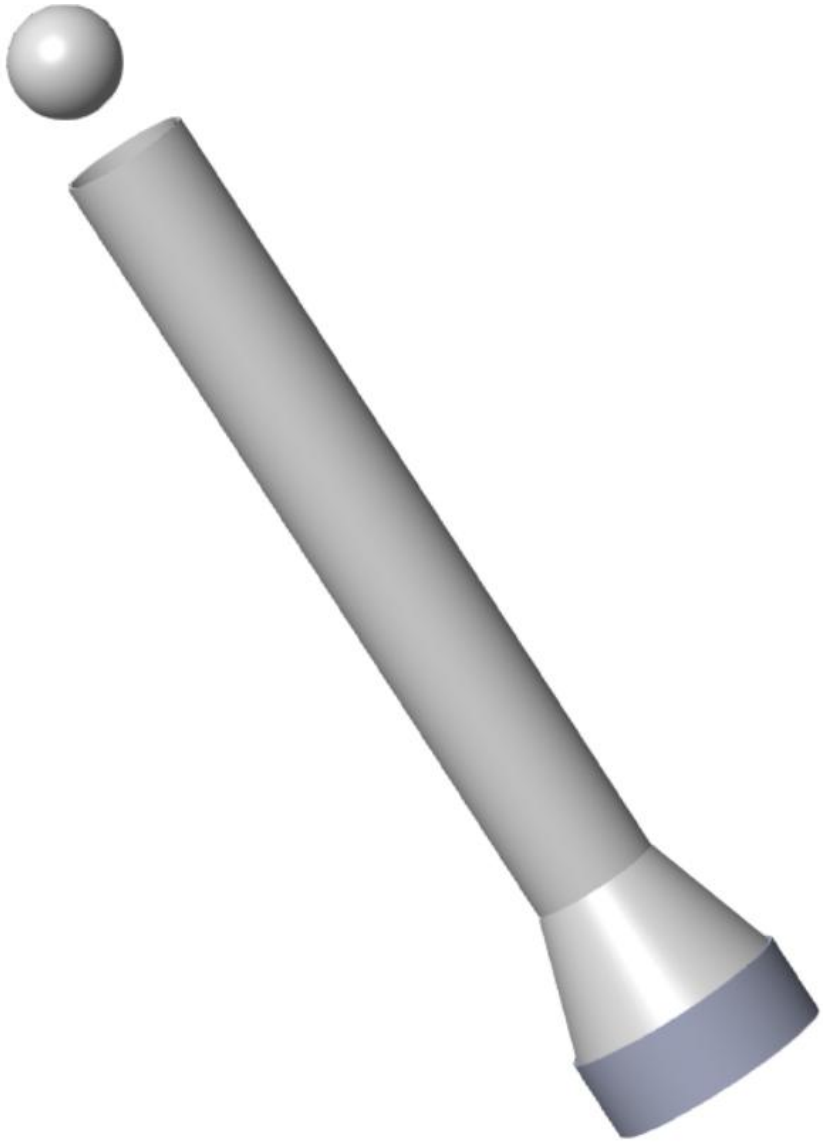


Figure : Isometric CAD view of the Ball Levitation system.

COMPONENTS AND MATERIALS

ELECTRONICS AND SENSORS

Component	Quantity	Purpose
Arduino Uno	1	Main Microcontroller
HC-SR04 Ultrasonic Sensor	1	Ball height sensing
L298N Motor Driver Module	1	Blower fan speed control via PWM
Blower Fan (12V DC)	1	Aerodynamic lift generation
12.4V 2000mAh LiPo Battery	1	Main Power Source
Jumper Wires Set	1	Interconnection Wiring

MECHANICAL PARTS

COMPONENT	QUANTITY	PURPOSE
Acrylic Transparent Tube (55mm ID, 300mm length)	1	Ball levitation channel
Cardboard sheets	-	Main frame
Hollow Ball 45mm diameter	1	Levitation object
Nozzle	1	Directs airflow into tube base
Base plate using Cardboard	1	Structural mounting frame

ELECTRICAL INTERFACING

The HC-SR04 ultrasonic sensor measures the distance to the ball by emitting and receiving ultrasonic pulses. The sensor operates on 5V and communicates with the Arduino Uno through two dedicated digital pins:

- **VCC** → 5V (Arduino)
- **GND** → GND (Arduino)
- **TRIG** → D9 of Arduino Uno (trigger pulse output)
- **ECHO** → D10 of Arduino Uno (echo signal input)

The Arduino computes the time-of-flight of the ultrasonic pulse to determine the ball's current height inside the tube and compares it against the desired setpoint for PID computation.

PWM Output:

- **D5** → L298N IN1 (speed control signal)
- **D6** → L298N IN2 (direction control signal)

The L298N motor driver receives the PWM signal from the Arduino and regulates the voltage supplied to the 12V blower fan accordingly. Higher PWM duty cycle increases fan speed and airflow, generating greater lift force on the ball.

Power Supply:

- The 12.4V LiPo battery supplies power to the L298N motor driver directly
- The L298N onboard 5V regulator supplies logic power to the Arduino Uno
- A 1N4007 diode is connected across the fan terminals for flyback protection against back EMF during motor switching

Connection Flow:

LiPo Battery → L298N Motor Driver → Blower Fan

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Arduino Uno (PWM D5/D6) → L298N → Fan speed control

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HC-SR04 Sensor (D9/D10) → Arduino → PID computation

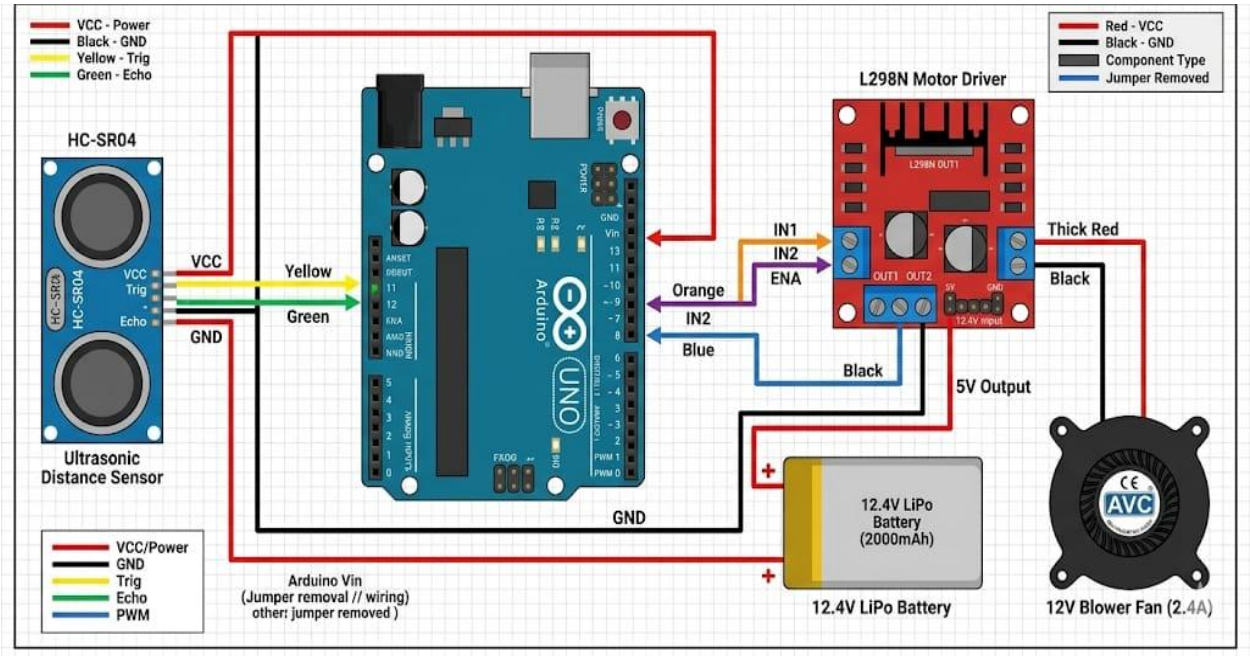


Figure : Wiring diagram for the Arduino-based prototype.

WORKING PRINCIPLE

The system utilizes closed-loop feedback control to stabilize the inherently unstable equilibrium between aerodynamic lift and gravity. An HC-SR04 ultrasonic sensor provides continuous positional data by measuring the real-time distance to the ball inside the transparent tube, allowing the Arduino Uno to detect any deviation from the target levitation height. The microcontroller processes this error via a PID algorithm to calculate the precise correction required in real-time.

This correction is output as a PWM signal to the L298N motor driver, which modulates the 12V supply voltage delivered to the blower fan. By varying the PWM duty cycle, the system controls the rotational speed of the fan and consequently the velocity of airflow entering the base of the tube through the nozzle. A higher duty cycle increases airflow and generates greater upward lift force on the ball, while a lower duty cycle reduces airflow and allows gravity to bring the ball downward.

The aerodynamic lift force acting on the 45 mm ball within the 55 mm diameter tube is highly sensitive to small changes in fan speed, making the system nonlinear in nature. The PID controller continuously accounts for this by combining proportional response to instantaneous error, integral action to eliminate steady-state offset, and derivative action to damp oscillations caused by sudden positional changes. Together these three terms ensure the ball remains suspended steadily at the desired height within the 300 mm tube.

A 1N4007 diode is connected across the fan motor terminals to dissipate inductive voltage spikes generated during PWM switching, protecting the L298N driver and Arduino from back EMF damage and ensuring smooth, stable fan operation throughout the levitation process.

MATHEMATICAL MODELLING

The airflow-based ball levitation system is inherently nonlinear due to the nature of aerodynamic lift force generation. The lift force acting on the ball is dependent on the fan airflow velocity, which itself has a nonlinear relationship with the PWM duty cycle supplied to the blower fan through the L298N motor driver.

1. Force Equation

Applying Newton's second law to the ball in the vertical direction, the governing equation of motion is:

$m * x''(t) = Flift(u) - m * g - Fdrag(x'(t))$

Where:

m = mass of the 45mm ball (kg)

x(t) = vertical position of the ball from tube base (m)

x'(t) = velocity of the ball (m/s)

x''(t) = acceleration of the ball (m/s^2)

g = acceleration due to gravity = 9.81 m/s^2

Flift(u) = aerodynamic lift force as a function of PWM input u

Fdrag = viscous drag force opposing ball motion

u = PWM duty cycle (0 to 255)

The aerodynamic lift force is modelled as proportional to the square of the airflow velocity, which is itself proportional to the fan PWM input:

$Flift(u) = K * u^2$

Where K is the aerodynamic coupling constant dependent on fan geometry, nozzle diameter, tube cross-section, and ball size.

The simplified equation of motion becomes:

$m * x''(t) = K * u^2 - m * g$

2. Linearization

Since PID controllers are linear, the system is linearized using a Taylor series expansion around the equilibrium operating point defined by equilibrium position x0 and equilibrium PWM input u0, where the ball is stationary and lift equals gravity:

At equilibrium: $K * u0^2 = m * g$

Define small perturbations:

delta x = x - x0 (small deviation in position)

delta u = u - u0 (small deviation in PWM input)

Expanding Flift around u0 and retaining only first-order terms:

$Flift(u0 + delta\ u) = K * u0^2 + 2 * K * u0 * delta_u$

The linearized equation of motion in terms of perturbation variables becomes:

$m * delta\ x''(t) = 2 * K * u0 * delta\ u(t)$

Defining the linearized actuator gain as:

$B = 2 * K * u0 / m$

The linearized plant equation is:

$delta\ x''(t) = B * delta\ u(t)$

3. Transfer Function

Applying the Laplace transform to the linearized equation assuming zero initial conditions:

$s^2 * X(s) = B * U(s)$

The open-loop transfer function G(s) relating ball position to PWM input is:

$G(s) = X(s) / U(s) = B / s^2$

Where:

X(s) = Laplace transform of delta x(t)

U(s) = Laplace transform of delta u(t)

B = linearized actuator gain = 2 * K * u0 / m

s = complex Laplace variable

This result is a double integrator, which represents a marginally stable open-loop system. Both poles lie at s = 0 on the imaginary axis, meaning any disturbance will cause the ball position to drift without bound without active feedback control. This is why the PID controller is essential to provide the necessary damping, restoring force, and steady-state correction.

4. PID Control Law

The PID controller adjusts the PWM output u(t) based on the positional error e(t) between the desired setpoint height r and the measured ball position x(t):

e(t) = r - x(t)

u(t) = u0 + Kp * e(t) + Ki * integral(e(t)dt) + Kd * de(t)/dt

Where u0 is the equilibrium feedforward PWM value that maintains hover at the setpoint in steady state, and Kp, Ki, Kd are the proportional, integral, and derivative gains respectively.

The feedforward term u0 can be estimated from the equilibrium condition:

u0 = sqrt(m * g / K)

Including u0 as a feedforward offset significantly reduces the burden on the integral term during initial capture, as the controller begins from a known approximation of the required steady-state fan speed rather than integrating up from zero.

When using a lightweight hollow ball, the aerodynamic coupling constant K is relatively high for a given tube diameter, meaning stable levitation can be achieved at moderate PWM values well within the L298N driver's operating range. However, the low ball mass also means the system responds rapidly to small PWM changes, making derivative gain Kd critical for preventing overshoot and sustained oscillation within the tube.

PID CONTROLLER DESIGN

Role of Each Term

In this system, the PID controller acts as the feedback mechanism that balances aerodynamic lift force against gravity. Because the system utilizes a lightweight hollow ball inside a confined tube, the controller must be specifically tuned to handle the nonlinear relationship between fan PWM duty cycle and the resulting airflow velocity, which produces a lift force that varies non-proportionally with fan speed.

- **Proportional (P):** Provides immediate correction based on the magnitude of positional error. If the ball drops below the setpoint, Kp increases the PWM duty cycle to raise fan speed and restore lift. While high Kp improves response speed, excessive gain will cause the ball to overshoot the setpoint and oscillate continuously within the tube due to the low inertia of the 45 mm ball.
- **Integral (I):** Eliminates steady-state error by accumulating past errors over time. It ensures the ball settles exactly at the target height rather than hovering slightly above or below it. This term effectively compensates for the specific weight of the ball and any consistent aerodynamic offset caused by the tube geometry and nozzle configuration.
- **Derivative (D):** Predicts future error based on the rate of change of position. This provides a braking effect as the ball approaches the setpoint from either direction. Kd is critical in this setup as it prevents the ball from overshooting and bouncing inside the tube by providing damping when the ball is moving rapidly toward or away from the desired levitation height.

Discrete Implementation (Arduino)

Since the Arduino Uno operates on a digital clock, the PID algorithm is implemented in discrete-time form that executes once per sampling interval Ts:

$e[k] = setpoint - measured_distance[k]$

$integral[k] = integral[k-1] + e[k] \times Ts$

$derivative[k] = (e[k] - e[k-1]) / Ts$

$u[k] = (Kp \times e[k]) + (Ki \times integral[k]) + (Kd \times derivative[k])$

Where,

$e[k]$ = position error at current sample k

$e[k-1]$ = position error at previous sample k-1

$integral[k]$ = accumulated error sum up to sample k

$derivative[k]$ = rate of change of error between samples

$u[k]$ = PID output at sample k

Ts = sampling interval in seconds

Kp, Ki, Kd = proportional, integral, derivative gains

The output u[k] is then clamped and mapped to a PWM value between 0 and 255 using:

$PWM\ output = constrain(u[k], 0, 255)$

This signal is sent from the Arduino to the L298N motor driver, which regulates the average voltage supplied to the blower fan, controlling its rotational speed and the resulting airflow lift force on the ball.

Tuning Method

Tuning for a lightweight ball in a confined tube requires an incremental experimental approach. The following sequence is recommended to ensure stability:

1. **Isolate Proportional (P):** Set Ki and Kd to zero. Slowly increase Kp until the ball begins to rise and react to changes in fan speed. The ball will oscillate around the setpoint at this stage.
2. **Add Damping (D):** Increase Kd to settle rapid oscillations. The goal is to reach a state where the ball feels viscous in the airflow rather than bouncy, slowing down smoothly as it approaches the setpoint.
3. **Introduce Integral (I):** Once the ball is hovering consistently near the setpoint, introduce a small Ki value to pull it precisely to the desired height and eliminate any remaining steady-state offset.

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Performance Observation Criteria

- Oscillation: Rapid up and down vibration of the ball indicates a Kp value that is too high. Reduce Kp in small steps until oscillation subsides.
- Settling Time: If the ball takes too long to return to the setpoint after a disturbance such as a gentle push, Kd or Kp requires upward adjustment to improve response speed.
- Fan Saturation: If u[k] consistently hits the PWM limit of 255 without the ball reaching the setpoint, the blower fan may be undersized for the ball weight or the tube geometry requires adjustment.
- Sensor Noise: High frequency jitter in the HC-SR04 readings can amplify the derivative term and cause erratic fan speed changes. A simple moving average filter on the sensor readings is recommended before feeding distance values into the PID computation.

VALUES OF PID CONSTANTS:

- $K_p = 6.0$
- $K_i = 0.01$
- $K_d = 3.5$

SIMULINK CLOSED-LOOP STUDY

The Simulink block diagram represents the levitation control loop in which the measured state — ball position within the tube — is fed back through the PID controller, passed through a saturation block limiting the PWM output to the range 0–255 (corresponding to 0–12V fan supply), and applied to the nonlinear plant model representing the aerodynamic and gravitational dynamics of the ball inside the 55 mm diameter, 300 mm long tube. The monitored outputs are the fan drive voltage, ball position, ball velocity, and PID control effort. The response plot shows a well-behaved transient for the selected operating region around the target levitation height.

The behaviour of the system during a simulated step input from rest to the desired setpoint height is analysed below:

- **Position:** The ball position begins with a non-zero deviation from the setpoint as the fan spins up from rest. It moves rapidly upward toward the equilibrium height, crosses slightly above the target due to aerodynamic overshoot, and finally settles at the desired levitation height as the PID integral term eliminates the remaining offset. The settling behaviour confirms effective closed-loop regulation within the physical constraints of the tube length.
- **Velocity:** The ball velocity exhibits an initial sharp positive transient as the blower fan generates sufficient lift to overcome gravity and accelerate the ball upward. The velocity then decays toward zero as the ball approaches the setpoint, indicating that the derivative term is effectively removing kinetic energy from the system and preventing continuous oscillation inside the tube.
- **Fan Drive Voltage:** The control voltage response corresponding to the PWM duty cycle is high at the start to overcome gravity and accelerate the ball from its rest position at the tube base. It decreases progressively once the ball approaches the setpoint and reduces further to the steady-state equilibrium voltage as the ball stabilizes. This behaviour is consistent with a stable closed-loop system that does not require sustained aggressive actuation at equilibrium, confirming correct PID gain selection.
- **PID Control Effort:** The computed PID output u[k] shows a large initial magnitude during the corrective phase when positional error is greatest. As

the ball approaches and settles at the setpoint, the control effort reduces to a low steady-state value sufficient only to maintain the equilibrium airflow against gravity, with negligible integral windup observed due to the saturation clamping applied at the PWM output stage.

EXPERIMENTAL IMPLEMENTATION

The project development includes a physical prototype demonstration, which confirms that the work progressed beyond theoretical software modelling into a realized electromechanical system. For reliable operation, the system requires precise physical alignment and a robust power electronic interface to manage the PWM-driven fan speed control and accurate ultrasonic position sensing within the confined tube geometry.

Assembly and Mechanical Alignment

In this vertically arranged system, mechanical alignment is critical for achieving a stable aerodynamic equilibrium within the tube.

- **Frame Construction:** The blower fan and nozzle assembly are mounted at the base of the structure, directing airflow vertically upward into the bottom opening of the 300 mm transparent tube. The tube is held rigidly in a vertical orientation above the nozzle to ensure a consistent and symmetric airflow column throughout its 55 mm inner diameter.
- **Sensor Positioning:** The HC-SR04 ultrasonic sensor is mounted at the top of the tube opening, facing downward into the tube interior. Its alignment must ensure it accurately tracks the ball's vertical position without acoustic obstruction from the tube walls, and its trigger and echo pins must be free of physical interference from the fan vibration transmitted through the frame.
- **Ball Centering:** The 45 mm ball must remain centered within the 55 mm tube to ensure aerodynamic lift acts symmetrically along the vertical axis. The small 5 mm radial clearance between the ball and tube wall acts as a natural guide, but any tilt in the tube will introduce consistent lateral contact friction that biases the ball position and degrades PID performance.
- **Structural Rigidity:** A rigid mechanical base is utilized to minimize vibrations transmitted from the blower fan motor to the tube and sensor assembly. Fan-induced vibration is a significant source of positional noise

in this system and must be mechanically isolated as far as possible to maintain clean sensor readings.

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Electronic Interfacing and Driver Selection

The control hardware is centered around the Arduino Uno, which performs sensor acquisition and generates the PWM command to regulate fan speed.

- **Driver Stage:** The L298N motor driver module is used as the power driver to control the voltage delivered to the 12V blower fan. The L298N provides adequate current capacity for the fan load and accepts the Arduino's 5V PWM logic signal directly without additional level shifting, making it well suited for this prototyping application.
- **Power Management:** The 12.4V LiPo battery supplies the L298N motor driver directly. The L298N's onboard 5V regulator steps down the supply voltage to provide a stable 5V power rail for the Arduino Uno and HC-SR04 sensor, eliminating the need for a separate voltage regulator and simplifying the power distribution architecture.
- **Grounding Scheme:** All electrical grounds including the Arduino Uno, HC-SR04 sensor, L298N motor driver, and LiPo battery negative terminal are tied together at a common ground point to ensure a valid and shared control reference. Poor grounding in this system can introduce offset voltages that corrupt the ultrasonic timing measurement and produce erroneous distance readings fed into the PID loop.

Factors for Reliable Operation

The likelihood of successful and stable levitation improves significantly when the following practical engineering considerations are met:

- **Back EMF Protection:** A 1N4007 flyback diode is connected across the blower fan motor terminals to dissipate inductive voltage spikes generated during PWM switching transitions. This protects the L298N driver output stage and Arduino PWM pins from transient overvoltage damage during high-frequency duty cycle changes.

- **Acoustic Isolation of Sensor:** The HC-SR04 sensor must not be mounted in direct contact with the fan motor housing or tube frame, as fan vibration conducted through the structure generates false echo triggers. Mounting the sensor on a soft rubber pad or isolating it with foam reduces this interference significantly.
- **Signal Integrity:** Clean grounding and stable power supply filtering are necessary to ensure the HC-SR04 echo timing signal remains free of electrical noise. Capacitive decoupling of the sensor VCC line with a 100nF ceramic capacitor placed close to the sensor power pins is recommended to suppress high-frequency switching noise from the L298N motor driver.
- **PWM Frequency Selection:** The Arduino's default PWM frequency on pins D5 and D6 is approximately 980 Hz, which is audible and can cause the fan motor to emit a tonal hum during operation. Increasing the PWM frequency using Timer1 register configuration reduces audible noise and improves the smoothness of fan speed modulation, which in turn reduces aerodynamic turbulence fluctuations acting on the ball

EXPERIMENTAL PROCEDURE

The experimental phase follows a systematic approach to transition from the Simulink model to a functional physical prototype. The procedure is divided into hardware verification, sensor calibration, and controller tuning.

Hardware Integration and Safety Check

1. **Wiring Verification:** Connect the Arduino Uno, HC-SR04 ultrasonic sensor, and L298N motor driver according to the wiring diagram. Verify that the TRIG and ECHO pins are connected to D9 and D10 respectively, and that the L298N IN1 and IN2 pins are connected to D5 and D6.
2. **Common Ground:** Ensure all components including the Arduino Uno, HC-SR04, L298N, and LiPo battery negative terminal share a common ground to prevent signal floating and reference voltage errors.
3. **Back EMF Protection:** Double-check the orientation of the 1N4007 diode across the blower fan motor terminals with the cathode connected toward the

positive supply rail to ensure correct flyback suppression during PWM switching.

4. **Power Verification:** Confirm that the L298N onboard 5V regulator is supplying a stable 5V to the Arduino Uno and HC-SR04 before connecting all logic pins and uploading the control sketch.

Sensor Calibration

1. **Range Mapping:** Using a simple serial print sketch, record the distance values reported by the HC-SR04 at known physical positions of the ball within the tube, for example at 5 cm, 10 cm, 15 cm, 20 cm, and 25 cm from the sensor.
2. **Offset Correction:** Compare the measured values against a ruler to identify any fixed offset introduced by the sensor mounting position at the top of the tube. Apply a constant correction value in the Arduino code to align reported distance with actual ball height.
3. **Noise Filtering:** Apply a software-based moving average filter to the HC-SR04 distance readings to reduce high-frequency jitter caused by acoustic reflections from the tube walls and fan vibration before feeding the value into the PID computation.

Controller Tuning and Deployment

1. **Open-Loop Test:** Run the blower fan at a fixed mid-range PWM value such as 150 out of 255 and observe whether the fan generates sufficient airflow to lift the ball partway up the tube, confirming the actuator and driver are functioning correctly.
2. **Initial PD Tuning:** Begin with Ki set to zero. Increase Kp until the ball rises and oscillates around the setpoint, then increase Kd to damp the oscillations and produce smooth settling behaviour.
3. **PID Implementation:** Once stable PD behaviour is achieved, introduce a small Ki value to eliminate any remaining steady-state offset between the ball's resting position and the desired setpoint height.
4. **Capture Routine:** Hold the ball manually at approximately the midpoint of the tube, around 15 cm from the sensor. Start the control loop and slowly release the ball once the fan begins modulating speed in response to positional

error. Observe the settling behaviour and reduce Kd if excessive oscillation occurs.

Data Acquisition

1. **Serial Plotting:** Use the Arduino Serial Plotter to monitor real-time ball position, error signal, and PWM output simultaneously during levitation to assess transient and steady-state behaviour visually.
2. **Stability Testing:** Gently push the levitating ball upward or downward by hand and release it to observe the system's disturbance rejection, specifically its ability to return the ball to the setpoint after a physical nudge without sustained oscillation.
3. **Fan Saturation Check:** Monitor the PWM output value during steady-state levitation to confirm it is not consistently saturating at 255, which would indicate insufficient fan capacity for the ball weight or tube configuration.

ADVANTAGES, LIMITATIONS AND FUTURE IMPROVEMENTS

The development and testing of the airflow-based ball levitation prototype provided insights into the strengths of the current design and identified specific areas where the system could be enhanced for better precision and stability.

1. Advantages

- Non-Contact Operation: Since there is no physical contact between the fan actuator and the ball during levitation, the system eliminates mechanical friction and wear on the ball surface. This makes airflow levitation particularly suitable for applications involving fragile or sensitive objects that cannot tolerate contact forces.
- Low-Cost Prototyping: By using an Arduino Uno and off-the-shelf components such as the L298N motor driver, HC-SR04 ultrasonic sensor, and a standard 12V blower fan, the project demonstrates that high-level control problems can be solved with an affordable and easily sourced hardware stack well within a student budget.
- High-Speed Response: The discrete sampling rate of the Arduino Uno allows for rapid PID corrections, which is essential for maintaining the unstable aerodynamic equilibrium. The lightweight nature of the 45 mm ball means the system responds quickly to small changes in fan speed, making fast closed-loop correction both necessary and achievable.
- Scalability: The PID control logic implemented here is fundamentally the same as that used in industrial applications such as pneumatic conveying systems, wind tunnel levitation rigs, and aerodynamic test platforms. The same principles scale directly to more complex multi-axis airflow control problems.

- Safe and Accessible: Unlike magnetic levitation systems that involve high currents through electromagnetic coils, airflow levitation operates with standard motor driver voltages and poses minimal risk of electrical hazard, making it a safer and more accessible platform for educational demonstration.

2. Limitations

- Nonlinearity and Capture Range: The relationship between fan PWM duty cycle and the resulting lift force on the ball is highly nonlinear and varies with ball position within the tube. The PID controller is only effective within a narrow operating region around the setpoint. If the ball moves too far from the equilibrium height, the controller loses authority and the ball either falls or hits the top of the tube.
- Sensor Noise and Latency: The HC-SR04 ultrasonic sensor is susceptible to acoustic reflections from the inner tube walls, multipath interference, and vibration-induced noise from the fan motor. These artefacts introduce jitter into the position measurement, which amplifies the derivative term of the PID controller and can cause erratic fan speed fluctuations if not filtered.
- Tube Geometry Constraints: The small clearance between the 45 mm ball and the 55 mm tube inner diameter creates a near-sealed aerodynamic channel. Minor misalignment of the ball or nozzle can create asymmetric airflow pressure around the ball, producing lateral forces that push the ball against the tube wall and increase friction unpredictably.
- Single Degree of Freedom: The current setup stabilizes the ball only in the vertical axis along the tube length. It does not account

for lateral movement or ball spin induced by asymmetric airflow, which can introduce consistent positional bias that the integral term alone may not fully compensate.

- Battery Discharge Effect: As the 12.4V LiPo battery discharges during operation, the available voltage to the L298N and blower fan decreases gradually. This changes the fan's speed-to-PWM characteristic over time, effectively shifting the system dynamics and requiring periodic recalibration of the PID setpoint or gain values.

3. Future Improvements

- Laser or Infrared Distance Sensing: Replacing the HC-SR04 ultrasonic sensor with a VL53L0X time-of-flight laser sensor would significantly reduce measurement latency and eliminate acoustic interference from the fan, enabling a tighter and more responsive control loop with reduced derivative noise.
- Adaptive PID or Gain Scheduling: Implementing a gain scheduling strategy where Kp, Ki, and Kd values are adjusted based on the ball's current position within the tube would compensate for the nonlinear lift-to-PWM relationship across the full 300 mm tube length, extending the effective capture range of the controller.
- Nozzle Flow Conditioning: Redesigning the air inlet nozzle with a flow straightener or honeycomb insert would produce a more uniform and laminar airflow column inside the tube, reducing turbulence-induced ball wobble and improving levitation stability at higher setpoint heights.
- Dual-Fan Actuation: Incorporating a second fan at the top of the tube blowing downward, operating in opposition to the base fan,

would provide bidirectional force authority over the ball. This would linearize the control problem and allow the ball to be held stably at any height within the tube with equal precision.

- Wireless Setpoint Control: Adding a Bluetooth module such as the HC-05 to the Arduino would allow the desired levitation height to be adjusted in real time from a mobile application, enabling dynamic setpoint tracking demonstrations and remote system monitoring without physical intervention.

CONCLUSION

The project successfully demonstrated the design, modelling, and control of an airflow-based ball levitation system. By applying PID closed-loop control to a Simscape Multibody simulation of the physical plant and subsequently implementing the controller on an Arduino Uno prototype, the team achieved stable levitation of a 45 mm ball at a target height of 150 mm within a 300 mm transparent tube. The transition from Simulink simulation to a physical prototype highlighted the critical importance of ultrasonic sensor calibration, acoustic noise filtering, and back EMF protection for reliable fan driver operation.

The system successfully demonstrated key principles of nonlinear control including aerodynamic instability, closed-loop feedback regulation, PID gain tuning, and real-time digital control implementation on embedded hardware. While the system faces practical limitations regarding tube geometry constraints, sensor acoustic noise, and battery discharge drift, it serves as a robust and accessible educational benchmark for understanding the control of unstable aerodynamic systems using low-cost embedded hardware. Future work incorporating time-of-flight laser sensing, adaptive gain scheduling, and dual-fan actuation would extend the system's precision, capture range, and operational stability significantly.